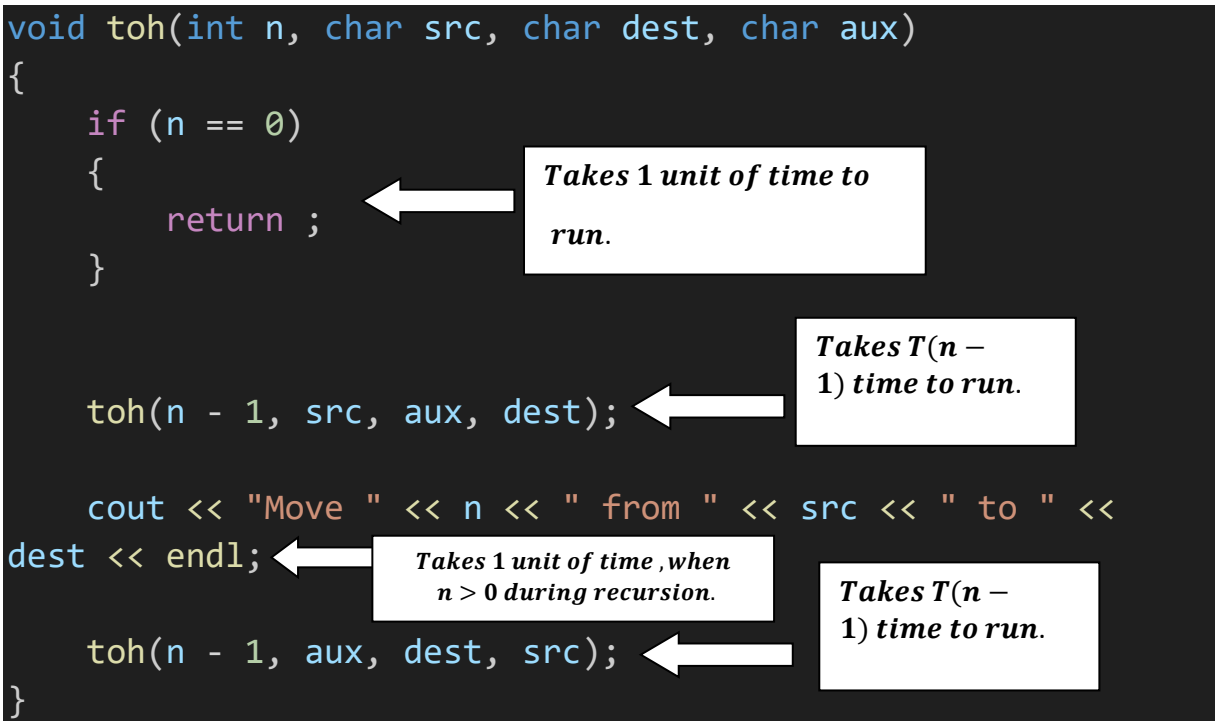
# *Tower Of Hanoi – Time Complexity through Substitution method.*

***From the given program we can say that :***




$To, start\ with\ , n > 0\ \ , TOH(n-1, src, aux, dest)$ $runs : T(n-1)\ times\ and\ TOH(n-1, aux, dest, src)$ $runs\ T(n-1) times.\ With\ Base\ case\ running\ at\ least\ once\ as\ recursion\ ends\ .$

$When\ n = 0\ , it\ runs\ constant\ time\ i.e\ T(0) = 1.$

***Hence, Base Case*:**

$$T(0) = 1$$

***And when*** $n > 0$ ,

$$T(n) = T(n-1) + T(n-1) + 1$$

$$i.e. T(n) = 2T(n-1) + 1$$

***Here*** $+1$ ***for printing the movement of the disk.***

```
    cout << "Move " << n << " from " << src << " to " <<
dest << endl;
```

***Hence we get the Recursive relation as*:**

$$T(n) = \begin{cases} 1 & , if\ n = 0 \\ 2T(n-1) + 1 & , if\ n > 0 \end{cases}$$

***Hence solving this recursive relation by substitution method we get*:**

$$T(n) = 2T(n-1) + 1$$

$$Now, T(n-1) = 2T(n-2) + 1$$

$$Substituting\ T(n-1) = 2T(n-2) + 1\ in\ T(n)\ we\ get:$$

$$T(n) = \ 2[2T(n-2) + 1] + 1$$

$$\Longrightarrow 2^2T(n-2) + 2 + 1$$

$$Again, T(n-2) = 2T(n-3) + 1$$

$$Substituting\ T(n-2) = 2T(n-3) + 1\ in\ T(n)\ we\ get:$$

$$T(n) = 2^2T(n-2) + 2 + 1$$

$$\Longrightarrow 2^2[2T(n-3) + 1] + 2 + 1$$

$$\Longrightarrow 2^3T(n-3) + 2^2 + 2 + 1$$

$$Now\ , if\ it\ runs\ upto\ `k`\ times\ we\ get:$$

$$T(n) = 2^kT(n-k) + 2^{k-1} + 2^{k-2} \ldots + 2^2 + 2 + 1$$

$$As, our\ base\ case\ is\ T(0) not\ T(1), hence:$$

$$n - k = 0\ , therefore\ , k = n\ , we\ get:$$

$$T(n) = 2^nT(n-n) + 2^{n-1} + 2^{n-2} + \cdots + 2^2 + 2 + 1$$

$$T(n) = 2^n T(0) + (2^{n-1} + 2^{n-2} + \cdots + 2^2 + 2 + 1)$$

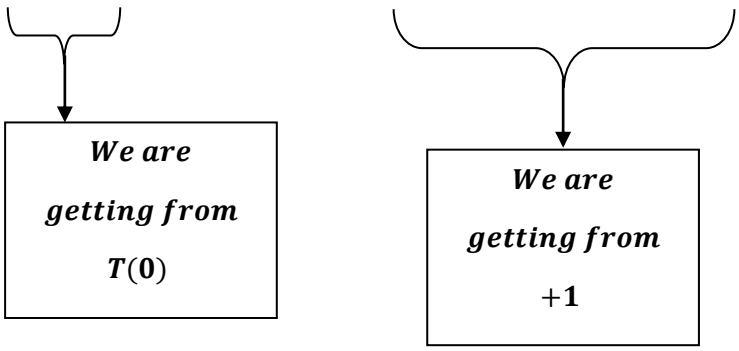
***Substituting*** $T(0) = 1$ ***, we get***:

$$\Longrightarrow 2^n \times 1 + 2^{n-1} + \cdots + 2 + 1$$

$$\Longrightarrow 2^n + (2^{n-1} + \cdots + 2 + 1)$$

***Now, conceptual scenario***:

$2^n$ + $(2^{n-1} + \cdots + 2 + 1)$




***Hence cannot combine both conceptually***:

***Geometric progression(series)upto`n`*** $\sum_{i=0}^{n} ar^i$ :

$$S(n) = ar^0 + ar^1 + ar^2 + \cdots + ar^n = a\left(\frac{r^{n+1}-1}{r-1}\right), where$$

$S(n)$ *is sum of first n terms ,a is coefficient and* $r$ *is common ratio between the consecutive terms.*

*Similarly,*

$$Geometric\ progression(series)upto`n-1` \sum_{i=0}^{n-1} ar^i :$$

$$S(n) = ar^0 + ar^1 + ar^2 + \cdots + ar^{n-1} = a\left(\frac{r^n-1}{r-1}\right), where$$

$S(n)$ *is sum of first n terms ,a is coefficient and* $r$ *is common ratio between the consecutive terms.*

*We get*:

$$\Rightarrow 2^n + \left\{1\left(\frac{2^n-1}{2-1}\right)\right\}$$

$$\Rightarrow 2^n + (2^n - 1)$$

$$\Rightarrow 2^{n+1} - 1$$

***putting Big O we get*:**

$\Rightarrow \boldsymbol{O}\ (\boldsymbol{2^{n+1}-1})$

$\Rightarrow \boldsymbol{O}(\boldsymbol{2^{n+1}}) - \boldsymbol{O}(\boldsymbol{1})$

$\Rightarrow \boldsymbol{O}(\boldsymbol{2^{n+1}})$

$\Rightarrow \boldsymbol{O}(\boldsymbol{2^{n}} \times \boldsymbol{2})$

$\Rightarrow \boldsymbol{O}(\boldsymbol{2^{n}}) \times \boldsymbol{O}(\boldsymbol{2})$

$\Rightarrow \boldsymbol{O}(\boldsymbol{2^{n}} \times \boldsymbol{2})$

$\Rightarrow \boldsymbol{O}(\boldsymbol{2^{n}})$.

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